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Covert feature-space adversarial perturbation using natural evolution strategies in distributed deep learning ☆

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ABSTRACT

Distributed Deep Learning (DDL) partitions deep neural networks across multiple devices, enhancing efficiency in large-scale inference tasks. However, this segmentation exposes intermediate-layer feature maps to new security vulnerabilities, expanding the attack surface beyond traditional input-level threats. This work investigates an adaptation of Natural Evolution Strategies (NES), named NES with Random Uniform Perturbation (NES-RUP), for adversarial manipulation of intermediate-layer feature maps in horizontally distributed inference systems. Instead of Gaussian-based perturbation sampling, the proposed method utilizes uniformly distributed noise and targets only a subset of feature map channels. This design improves stealth by using uniform noise distributions, which avoid extreme outliers and limit perturbations to bounded ranges, keeping activations closer to their clean values and thereby reducing anomaly detection likelihood, while also aligning with the performance and privacy constraints of AIoT-enabled smart environments. Extensive experiments on VGG16, ResNet50 and DeiT-Tiny (a Vision Transformer) using the CIFAR-10 and Mini-ImageNet datasets demonstrate that the adapted NES method achieves high misclassification rates with minimal feature-level distortion, preserving the statistical characteristics of natural feature activations. Furthermore, it successfully bypasses common defenses such as low-pass filtering and feature map anomaly detection (e.g., PseudoNet), revealing critical vulnerabilities in collaborative inference. These findings underscore the need for dedicated defense strategies that address intermediate-layer threats in secure AIoT infrastructures.

1. Introduction

The growing adoption of distributed deep learning (DDL) has transformed applications in Artificial Intelligence of Things (AIoT), edge computing, and autonomous systems, enabling smart societies where intelligent services operate collaboratively across resource-constrained devices. Unlike centralized architectures, where all computations are performed on a single powerful machine, DDL partitions models across multiple nodes, often distributing layers between edge and cloud components, facilitating decentralized and scalable inference [1–3]. Common DDL paradigms include horizontal collaboration, where a deep neural network is partitioned layer-wise across multiple devices and each device processes a subset of consecutive layers before forwarding the intermediate feature maps to the next node. Vertical collaboration refers to cases where different feature representations or portions of the input data are processed across multiple entities in parallel and later aggregated. Hybrid collaboration combines elements of both partitioning modes to balance efficiency and flexibility. This approach is particularly valuable for real-time decision-making, where computational efficiency and low latency are critical [4]. By distributing layers of a deep neural network across edge and cloud components, DDL allows resource-constrained devices to contribute to deep learning tasks without processing an entire model [5]. However, this partitioning creates new vulnerabilities, particularly during the inference phase, where adversaries can manipulate intermediate feature maps to degrade model performance while remaining more covert than traditional input-based attacks, posing significant security and privacy concerns in collaborative AIoT environments [6,7]. Fig. 1 illustrates a typical horizontally collaborated DDL system, where intermediate feature maps flow between devices until final classification is performed at the last node.

One of the primary vulnerabilities in distributed inference arises from feature map manipulation. In horizontally partitioned architectures, each node processes a subset of layers before transmitting

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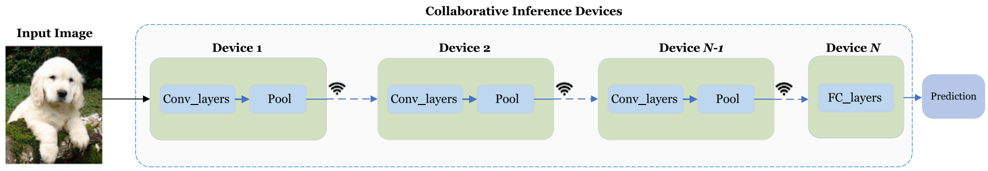

Fig. 1. Illustration of a horizontally collaborated distributed deep learning system, where a deep neural network is partitioned across multiple devices.

its intermediate representations to the next device. If an adversary compromises such a node, they can perturb feature maps to induce misclassification in subsequent layers. Numerous input-space attacks (e.g., FGSM [8], PGD [9], CW [10], DeepFool [11], JSMA [12], SparseFool [13]) have demonstrated the effectiveness of gradient-based perturbations in white-box settings, but these approaches assume full model access and therefore do not directly apply to distributed inference scenarios.

Building on the vulnerabilities inherent in distributed inference, adversaries often exploit black-box scenarios where model details remain hidden. In addition to advanced strategies such as query-based optimization or transferability, simple random-noise perturbations such as Gaussian or uniform injection are widely reported as standard black-box baselines in adversarial attack [14], and serve as non-optimized references against which stronger strategies can be evaluated. Natural Evolution Strategies (NES) is a prominent black-box approach that bypasses the need for direct gradient access by estimating directional changes through repeated queries, thereby remaining effective even when model information is limited [15]. This is particularly relevant for AIoT systems, where models are often split across heterogeneous devices and full-model access is infeasible. Although input-based attacks have been extensively studied, defenses such as adversarial training, input preprocessing, and feature squeezing offer only partial mitigation by filtering out adversarial noise at the input level. In contrast, feature map-based perturbations operate deeper within the network, beyond the input stage, where traditional defenses are less effective, leaving distributed inference pipelines particularly vulnerable to such adversarial manipulations.

Prior research on adversarial threats in distributed inference has primarily focused on input-level perturbations and edge-layer manipulations, where adversarial perturbations are introduced in the early stages of a partitioned model. Although intermediate-layer perturbations have also been investigated, existing approaches typically apply indiscriminate noise injection without optimization, reducing their stealthiness and increasing detectability. While NES has previously shown effectiveness in crafting input-space adversarial perturbations under black-box conditions, their applicability to intermediate-layer feature spaces within distributed inference scenarios remains largely unexplored. To bridge this gap, we propose a covert adversarial perturbation method, named NES-RUP (Natural Evolution Strategies with Random Uniform Perturbation), specifically adapted for intermediate-layer feature maps in distributed inference systems common in privacy-aware AIoT infrastructures.

Unlike traditional NES implementations, the proposed method employs uniformly distributed noise instead of Gaussian noise to sample perturbations. Specifically, uniform noise perturbations are bounded and therefore avoid extreme outliers, keeping perturbed activations statistically closer to their clean counterparts than Gaussian perturbations. This helps make anomalies introduced by the attack harder to detect by statistical or learning-based methods. Furthermore, rather than indiscriminately perturbing all feature channels, our approach selectively introduces adversarial perturbations into random subsets of feature channels, substantially reduces perturbation magnitude and computational overhead. This selective channel perturbation approach minimizes overall distortion of intermediate features, preserving normal feature statistics and further evading detection. By iteratively refining perturbations through query-based interactions with the targeted model, this NES-based approach induces effective misclassifications while closely preserving the statistical properties of natural activations, maintaining a low detection profile and significantly outperforming conventional intermediate-layer perturbation techniques.

The primary objective of this research is to evaluate whether our selective, NES-optimized intermediate-layer perturbations can achieve high misclassification rates while successfully evading existing detection and mitigation methods, such as filtering-based defenses and learned anomaly detection mechanisms. We rigorously validate the proposed adversarial perturbation method on popular convolutional neural networks, including VGG16 and ResNet50, using benchmark datasets CIFAR-10 and Mini-ImageNet. Our experiments demonstrate that selective, NES-optimized perturbations at intermediate layers effectively disrupt model predictions, inducing high rates of misclassification with minimal feature-level distortion. This study expands adversarial perturbation research within distributed inference settings, highlighting critical vulnerabilities of existing defenses and underscoring the urgent need for advanced detection and mitigation techniques specifically designed for intermediate-layer adversarial perturbations in distributed deep learning systems. Such techniques are essential for securing AI-driven services in smart society infrastructures, where real-time, privacy-sensitive decisions rely on collaborative inference across heterogeneous IoT devices.

The contribution of this paper is the development of NES-RUP, a selective intermediate-layer adversarial perturbation method that combines Natural Evolution Strategies with bounded uniform noise to achieve strong misclassification with minimal feature-level distortion in horizontally partitioned deep learning. We show that NES-RUP outperforms non-optimized baselines such as RUP and GNA across VGG16, ResNet50, and DeiT-Tiny models, while remaining highly stealthy due to its channel-selective perturbation design. We further demonstrate that NES-RUP can bypass lightweight defenses, including low-pass filtering and PseudoNet-based anomaly detection, revealing significant vulnerabilities in distributed inference pipelines. Finally, we support our analysis with qualitative feature-maps and attention-based visualizations to illustrate how optimized and non-optimized perturbations alter internal representations.

The structure of this paper is as follows. Section 2 reviews the background and related work. Section 3 introduces the system model and assumptions, outlining the distributed inference framework, adversarial threat models, and scenarios considered in this study. Section 4 presents the proposed approach, detailing the NES-driven adversarial perturbation method. Section 5 describes the experimental methodology, including datasets, model architectures, evaluation metrics, and baseline defenses. Section 6 reports the results and provides a detailed analysis, comparing the effectiveness, stealth, and robustness of different attack and defense strategies. Finally, Section 7 concludes the paper by summarizing key findings and outlining directions for future research.

2. Background and related work

DDL has gained traction as a means to manage the computational complexity of deep neural networks (DNNs), particularly in AIoT-enabled systems, where resource-constrained devices collaboratively process large-scale data to support real-time, intelligent services [16, 17]. This is often facilitated through architectures like parameter-server frameworks, where a central server aggregates updates from distributed nodes, or decentralized systems such as gossip-based setups, where nodes exchange information directly to enhance flexibility [18,19].

In particular, distributed inference where the execution of a trained model across multiple devices has become increasingly common in edge-cloud settings or IoT networks with resource-constrained devices [20,21]. For example, the first few layers of a convolutional neural network (CNN) may run on an edge device, while subsequent layers execute on a more powerful server. Alternatively, different parts of the input data might be processed on separate devices and their results combined [22]. Such partitioning not only saves local computation or bandwidth but also helps preserve privacy by exposing only partial data or model segments to each node [23]. However, these same distributed structures also expand the attack surface, as each intermediate device handling part of the inference pipeline becomes a potential point of compromise [24]. Communication overhead between nodes introduces latency and potential failure points, while the distributed structure creates opportunities for adversaries to tamper with intermediate feature representations, making covert adversarial perturbations in distributed inference an increasingly critical concern. Studies underscore that such vulnerabilities amplify the risk of undetected manipulations, necessitating a deeper exploration of security in these intermediate stages [25].

In centralized deep learning, adversarial perturbations are extensively studied, with input-level attacks such as FGSM, PGD, CW, DeepFool, JSMA, and SparseFool [8–13] demonstrating how gradient-based methods can force misclassification under white-box access. Black-box techniques like NES [15,26] further bypass gradient requirements using query-based optimization. While highly effective in centralized settings, these methods assume a unified model and do not directly apply to distributed inference pipelines where intermediate-layer feature maps are exposed across devices. Although these techniques primarily target centralized models, input perturbations remain possible in distributed settings (e.g., FGSM, PGD, or random-noise attacks such as GNA [27]). Even with only partial access to the first few layers, adversaries can still craft perturbations that transfer across unknown downstream layers [28,29]. However, input-level attacks offer limited stealth, motivating the need for intermediate feature-space manipulation.

Beyond input-level attacks, the distribution of inference across multiple devices introduces distinct threat models, with intermediate feature-space attacks emerging as a critical concern [30,31]. Research on adversarial perturbation in distributed inference has explored various attack surfaces, including edge-layer perturbations and deeper middle-layer manipulations. A recent study demonstrated the feasibility of edge-only universal adversarial perturbations (UAPs), where an adversary with access solely to the initial layers crafts perturbations that transfer to the cloud portion of the model, causing misclassification [32]. Their approach relies on learning class-specific features from edge-layer outputs and optimizing UAPs accordingly. However, edge-layer perturbations primarily manipulate early-stage feature representations, which are less semantically meaningful, potentially limiting their stealth and effectiveness in later stages. In contrast, intermediate-layer attacks operate deeper within the feature hierarchy, where representations become more abstract and semantically refined. When a model is partitioned and intermediate feature maps are computed or transmitted across multiple devices, an adversary controlling a compromised node can tamper with these tensors. By intercepting and altering feature maps at partition points, adversaries can influence final classification outcomes even with only partial knowledge of the downstream layers [29]. [27] showed that perturbing intermediate activations in horizontally collaborated inference can significantly degrade accuracy. We assume an adversary with control over a single intermediate node and black-box access to downstream model outputs (e.g., prediction labels). Although NES has been widely adopted for pixel-level adversarial attacks, its extension to intermediate feature maps in horizontally partitioned inference remains largely unexplored. Feature activations differ fundamentally from raw pixels: they follow non-Gaussian statistics, are transmitted across distributed devices, and must be perturbed in a way that both preserves stealth and respects resource constraints. To address these challenges, our design departs from prior NES usage in two key ways. First, we employ uniformly distributed perturbations, which introduce bounded changes that better align with the natural variability of feature activations compared to Gaussian noise. Second, we introduce channel subsampling, reducing perturbation footprint, improving query efficiency, and matching the bandwidth limitations of AIoT systems. These modifications are motivated by distributed inference constraints to achieve a superior success-to-distortion trade-off compared to Gaussian-based NES.

Despite ongoing research, critical gaps remain in adversarial robustness for distributed inference. Training-phase threats such as data poisoning or gradient tampering [30,33,34] differ fundamentally from inference-time manipulations, which can immediately alter predictions. Intermediate-layer perturbation studies [27] show the vulnerability of horizontally partitioned CNNs, but they typically rely on random or Gaussian noise, limiting both stealth and efficiency. While NES-based adversarial attacks have been explored in the input space [35–37], to our knowledge no prior work has studied channel-specific uniform-noise perturbations for intermediate feature maps in distributed inference. By selectively targeting critical feature channels with bounded uniform perturbations, NES-RUP enhances both adversarial impact and undetectability, directly motivating the contributions of this paper.

Given the vulnerabilities in inference-time attacks, defenses against adversarial attacks in DDL serve as a benchmark for evaluating our covert intermediate feature-space attack resilience, with several strategies directly applicable to feature map security. Robust aggregation schemes like Krum and trimmed mean methods filter outlier updates during training, but their focus on gradient aggregation offers limited protection against inference-time feature map tampering [38,39]. Node reputation and anomaly detection, as explored by [40], monitor update credibility, yet their effectiveness wanes against stealthy perturbations mimicking natural activations, a key challenge our attack addresses. Specific to feature maps, [27] employ low-pass filtering to mitigate noise, recovering accuracy, while [41] propose SHEATH, a framework designed to detect adversarial noise in horizontal collaboration using PseudoNet. We assess the resilience of our attack against filtering-based defenses and anomaly detection mechanisms, revealing critical blind spots in existing countermeasures. Adversarial training, incorporating perturbed feature maps into training, enhances robustness but struggles with computational overhead and generalization [42]. These defenses, while promising, often assume broader noise patterns, leaving gaps our selective, channel-focused attack exploits, necessitating more tailored countermeasures.

This paper proposes NES-RUP, a covert adversarial perturbation method targeting intermediate-layer feature map channels in distributed inference, designed to degrade model performance while evading detection. NES-RUP employs an adapted NES optimization strategy that selectively perturbs only a subset of feature channels using uniformly distributed noise, rather than applying indiscriminate Gaussian noise. By refining perturbations iteratively under black-box constraints, our method achieves high stealth and efficiency, preserving the statistical similarity of activations while significantly degrading classification accuracy. Unlike prior NES-based attacks that operate in input space or inject unoptimized random noise at intermediate layers, NES-RUP is explicitly tailored to the feature space of horizontally partitioned inference and uses two-sided NES with channel-selective uniform perturbations, enhancing stealth and query efficiency under AIoT bandwidth and detectability constraints.

We evaluate NES-RUP on VGG16 [43] and ResNet50 [44], using CIFAR-10 and Mini-ImageNet datasets. Experimental results demonstrate that NES-RUP can bypasses defenses such as low-pass filtering and PseudoNet, exposing critical vulnerabilities in distributed inference pipelines. These findings reveal persistent blind spots in existing defense mechanisms and underscore the urgent need for lightweight, distributed detection strategies to secure AIoT-enabled smart society infrastructures against stealthy inference-time attacks.

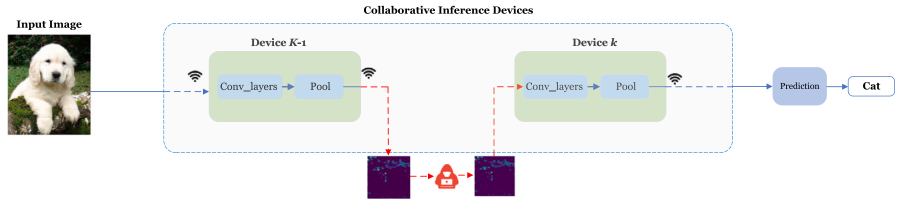

Fig. 2. An adversary at Device $K-1$ injects adversarial perturbations into feature maps before they reach Device K in a horizontally partitioned DDL setup. The perturbations are optimized via black-box queries to induce misclassification while maintaining stealth.

3. System model and assumptions

DDL enables large-scale inference by partitioning models across multiple computational nodes, where each node processes a subset of layers before passing intermediate feature maps to the next device. This framework is particularly beneficial in edge-cloud environments, AIoT systems, and resource-constrained devices, where deploying full deep learning models on a single device is impractical. However, model partitioning not only enhances scalability but also introduces security risks, as intermediate feature maps become vulnerable to adversarial manipulation. In this section, we define the distributed inference framework, outline our adversarial threat model, describe the mechanics of feature map perturbation, and discuss system-level constraints that influence attack feasibility.

3.1. Distributed deep learning framework

Modern DDL architectures employ horizontal collaboration, where a deep learning model is partitioned across multiple computational nodes. Each node is responsible for processing a subset of the network's layers and transmits intermediate feature maps to subsequent devices in the inference pipeline. This approach is widely used in resource-constrained environments such as edge computing, IoT, and federated learning, where centralized processing is infeasible due to latency, power constraints, or privacy concerns. While this distributed architecture enhances scalability and efficiency, it also introduces security vulnerabilities that adversaries can exploit to manipulate the model's inference process.

A model such as VGG16 can be deployed across multiple devices, with each device processing a subset of layers. The system follows a sequential partitioning where each node receives an input from the previous node, processes its assigned layers, and forwards the transformed feature maps to the next device. This structure ensures efficient utilization of computational resources while maintaining privacy by limiting access to raw input data. However, this structure also introduces vulnerabilities at partition boundaries, where adversaries can manipulate feature maps before they propagate through the remaining network.

3.2. Threat model

We assume an adversary who gains control over a single intermediate device within the DDL inference pipeline. Unlike conventional attacks that manipulate raw inputs, this adversary operates exclusively in the feature space by perturbing intermediate activations. The goal is to induce misclassification at the final output while remaining undetected by standard input preprocessing defenses. The attack follows a partial knowledge threat model. The adversary can observe and manipulate the feature maps at the compromised node but lacks access to model weights, gradients, or architectural details beyond that point. However, the adversary has black-box query access to the downstream model specifically, they can observe classification outputs (e.g., predicted labels or confidence scores) in response to perturbed feature maps. Optimization proceeds by iteratively refining the perturbation based on output variations, without requiring access to internal gradients. This query-efficient approach is well-suited for AIoT environments where attackers cannot fully inspect or reverse-engineer the model. Fig. 2 illustrates the attack scenario, where an adversary at Device $K-1$ injects perturbations into transmitted feature maps to influence final predictions at Device K. By selectively perturbing a subset of feature channels and preserving the statistical properties of natural activations, the adversary can achieve high misclassification rates while minimizing detectability. Perturbation stops once a desired misclassification confidence is achieved, ensuring both efficiency and stealth.

Remark. In addition to intermediate feature-space manipulations, Trojan attacks pose another critical threat by directly compromising the model partition itself, modifying the code or hardware running a sequence of layers on a device. These attacks alter model parameters or computations, enabling adversaries to stealthily degrade or hijack inference outcomes [45]. Techniques such as scaling weights, injecting random noise, polarity inversion, or statistical replacements can induce misclassifications when triggered by specific inputs [46]. While Trojan attacks allow deeper, more persistent manipulation, they require extensive access to model execution environments. In contrast, our approach achieves stealthy feature-map perturbations at runtime without modifying model weights or architecture, making it a more practical and adaptable adversarial strategy for inference-time attacks.

It should be noted that the assumed adversary capabilities represent a strong yet realistic scenario. In distributed AIoT deployments, intermediate feature maps are often transmitted over untrusted or resource-constrained links (e.g., Wi-Fi, Bluetooth, or mesh IoT), where encryption and integrity checks may be disabled to reduce latency and energy consumption. Even when secure links are employed, an attacker compromising the intermediate device itself (through malware or firmware modification) can still manipulate feature maps before encryption is applied. These considerations ensure that the threat model reflects plausible adversarial scenarios in collaborative inference rather than relying on impractical assumptions.

Real deployments may reconfigure the split at runtime and exhibit timing jitter. Our threat model and NES-RUP remain applicable because the adversary perturbs whichever intermediate tensor is present at the current cut and the optimization uses only forward queries. Thus the method does not rely on a fixed partition or strict synchrony.

3.3. Feature map perturbation under system constraints

Feature map perturbation refers to the injection of adversarial noise at internal layers of a deep neural network, targeting intermediate representations during inference. Unlike input-space attacks that alter raw inputs, these manipulations remain hidden within the model, complicating detection. In distributed deep learning systems, where layers are partitioned across multiple devices, feature maps must be transmitted between nodes, exposing them as a new attack surface. If an adversary compromises an intermediate device, they can tamper with outgoing feature maps before they propagate to subsequent layers, thereby influencing final predictions without altering the original input.

This vulnerability is exacerbated by the communication overhead and potential insecurity of inter-device links. Without strong cryptographic protection, which may be impractical in real-time and resource-constrained AIoT environments, intermediate feature maps can be intercepted or modified during transmission. Furthermore, the limited computational resources of edge devices restrict the use of heavyweight defenses such as adversarial training, redundancy-based verification, or feature-space consistency checks. As a result, lightweight anomaly detection strategies, including filtering-based defenses and learned detectors such as PseudoNet, are typically deployed. While these methods mitigate coarse or indiscriminate perturbations, they remain vulnerable to more covert manipulations that preserve the statistical properties of natural feature activations.

The inherent stealth of feature-space perturbations further complicates defense. Operating on internal activations rather than visible inputs, these attacks bypass conventional input preprocessing and exploit the subtle variations of feature maps, especially when applied to only small subsets of channels. This combination of constrained environments, insecure communication, and the covert nature of perturbations highlights the need for advanced adversarial strategies and robust defenses tailored to distributed inference.

4. Proposed approach

DDL frameworks enable scalable inference by partitioning neural networks across multiple devices, exchanging intermediate feature maps during collaborative execution. While efficient, this architecture introduces security vulnerabilities, particularly at partition boundaries where adversaries can manipulate feature maps to induce misclassification. Existing perturbation strategies that indiscriminately modify all feature channels often require large distortions and are computationally expensive, increasing their detectability. Unlike indiscriminate intermediate-layer noise, NES-RUP is *channel-selective*, using uniformly sampled directions that reduce outliers and preserve the natural statistics of activations.

4.1. Baseline: Random channel perturbation

For baseline comparison, we implement two simple non-optimized adversarial perturbation methods: Gaussian Noise Attack (GNA) and Random Uniform Perturbation (RUP). Both operate under a black-box setting by randomly perturbing selected intermediate channels without optimization or gradient access. Gaussian noise is a widely used robustness baseline [27,47], while the robustness of classifiers to random uniform noise has also been formally analyzed [48].

Given an intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$ with C channels, the adversary first selects a random subset of N channels for potential perturbation:

$$C_{\text{rand}} = \{c_1, c_2, \dots, c_N\}, \quad c_i \sim \text{Uniform}(\{1, 2, \dots, C\}). \tag{1}$$

In each iteration t, a single channel $c_t \in C_{\text{rand}}$ is perturbed. For GNA, the perturbation is applied by adding zero-mean Gaussian noise:

$$F'_{c_t,:,:} = F_{c_t,:,:} + \eta \cdot \mathcal{N}(0, 1), \tag{2}$$

where $\mathcal{N}(0, 1)$ denotes standard Gaussian noise and η is a scaling factor controlling the perturbation magnitude. In contrast, RUP replaces the Gaussian noise with uniformly distributed noise sampled from $\mathcal{U}(0, 1)$:

$$F'_{c_t,:,:} = F_{c_t,:,:} + \eta \cdot \mathcal{U}(0, 1). \tag{3}$$

The perturbation process continues iteratively until the classification confidence for the true class drops below a predefined threshold, or a maximum number of steps is reached. While both GNA and RUP offer minimal, query-free adversarial setups, they lack structured optimization and fail to account for the importance of individual channels. Consequently, these baselines are less effective and more susceptible to detection in realistic AIoT inference pipelines.

4.2. NES-RUP optimized perturbation

While random channel perturbation provides simplicity, it lacks explicit optimization, resulting in inefficient and detectable perturbations. To address this limitation and further enhance the effectiveness and stealth of adversarial attacks, we develop NES-RUP, which integrates a gradient-free optimization method NES. NES is well-suited for horizontally distributed DDL systems, particularly in AIoT settings, where adversaries may only observe output predictions and lack access to gradients, weights, or internal states.

Specifically, NES treats the target model as an oracle, repeatedly sampling random perturbations and estimating gradient approximations based on variations in model confidence scores. Given an intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$, we aim to iteratively construct an adversarial perturbation tensor $\Delta \in \mathbb{R}^{C\times H\times W}$ using gradient approximations obtained via random sampling. At each iteration, we approximate the gradient of a classification loss $\mathcal{L}$ (typically cross-entropy or negative log-likelihood) with respect to the perturbation tensor as follows:

$$\hat{\nabla}_\Delta \mathcal{L} = \frac{1}{2\sigma K} \sum_{k=1}^{K} \left[\mathcal{L}(F + \Delta + \sigma u_k) - \mathcal{L}(F + \Delta - \sigma u_k)\right] u_k, \tag{4}$$

where u_k is sampled from a uniform distribution $\mathcal{U}(0, 1)$, K denotes the number of samples for estimating the gradient, and σ represents the perturbation exploration scale. Uniformly distributed noise introduces bounded perturbations that produce consistent magnitude changes across selected channels. Unlike Gaussian noise, which may include outlier values due to its unbounded tails, uniform noise ensures more stable perturbation patterns, better mimicking natural activation variations. Once the NES-based gradient estimate $\hat{\nabla}_\Delta \mathcal{L}$ is computed, the perturbation tensor is updated iteratively by normalizing and applying the gradient approximation:

$$\Delta \leftarrow \Delta + \alpha \cdot \frac{\hat{\nabla}_\Delta \mathcal{L}}{\|\hat{\nabla}_\Delta \mathcal{L}\|}, \tag{5}$$

where α is the step size determining the magnitude of each update. This iterative process continues until either (i) the classification confidence for the original label falls below a predefined threshold or (ii) a maximum iteration limit is reached. The attack also terminates if the model's top-1 prediction (i.e., the most probable output class) no longer matches the true label, indicating a successful misclassification. This two-sided approach allows for more accurate gradient estimation, improving convergence speed and attack efficiency.

By adopting NES optimization with uniformly distributed perturbations for intermediate-layer feature maps, our method achieves significantly enhanced stealth, lower perturbation magnitude, and improved efficiency compared to baseline random perturbation methods. The targeted NES-based perturbation method closely maintains statistical similarity to natural activations, making perturbations challenging to detect within distributed deep learning inference pipelines. This makes the attack especially dangerous in AIoT-enabled smart environments, where inference pipelines span untrusted and resource-constrained devices, and detection mechanisms must operate in real-time.

The pseudocode for the proposed NES-RUP method is presented in Algorithm 1. In the context of horizontally partitioned inference in AIoT-enabled systems, the input feature map F corresponds to the intermediate activations at the device under adversarial influence (e.g., Device 2 or Device 3), where perturbations are strategically applied.

Algorithm 1: NES-RUP: Natural Evolution Strategies with Random Uniform Perturbation

Input: Clean feature map $F \in \mathbb{R}^{C\times H\times W}$, model $\mathcal{M}$, label y, number of channels to perturb N, maximum steps T, NES step size α, noise scale σ, number of samples K, confidence threshold P_{th}

Output: Perturbed feature map $F_{\text{adv}} = F + \Delta$

Initialize $\Delta \leftarrow 0$, $C_{\text{total}} \leftarrow$ total number of channels in F
Randomly select N channel indices: $C_{\text{rand}} \subset \{1, \dots, C_{\text{total}}\}$;
Construct binary mask $M \in \{0,1\}^{C\times H\times W}$ with ones at selected channels C_{rand};
for $t \leftarrow 1$ **to** T **do**
  Estimate gradient: $\hat{\nabla}_\Delta \mathcal{L} \leftarrow 0$;
  for $k \leftarrow 1$ **to** K **do**
    Sample $u_k \sim \mathcal{U}(0,1)^{C\times H\times W}$;
    Mask: $u_k \leftarrow u_k \odot M$; `// Restrict perturbation to selected channels`
    Compute loss difference:
    $\ell^+ \leftarrow \mathcal{L}(\mathcal{M}(F + \Delta + \sigma u_k), y)$;
    $\ell^- \leftarrow \mathcal{L}(\mathcal{M}(F + \Delta - \sigma u_k), y)$;
    $\hat{\nabla}_\Delta \mathcal{L} \leftarrow \hat{\nabla}_\Delta \mathcal{L} + (\ell^+ - \ell^-) \cdot u_k$;
  $\hat{\nabla}_\Delta \mathcal{L} \leftarrow \frac{1}{2\sigma K} \hat{\nabla}_\Delta \mathcal{L}$;
  Normalize gradient: $\hat{\nabla}_\Delta \mathcal{L} \leftarrow \frac{\hat{\nabla}_\Delta \mathcal{L}}{\|\hat{\nabla}_\Delta \mathcal{L}\|}$;
  Update perturbation: $\Delta \leftarrow \Delta + \alpha \cdot \hat{\nabla}_\Delta \mathcal{L}$;
  if *confidence*$(\mathcal{M}(F + \Delta), y) < P_{th}$ ***or*** $\arg\max \mathcal{M}(F + \Delta) \neq y$ **then**
    break
Return $F_{\text{adv}} \leftarrow F + \Delta$

4.3. Practical considerations and implementation

In horizontally distributed deep learning systems, particularly those deployed in AIoT-enabled smart environments, practical adversarial strategies must account for resource constraints, bandwidth limitations, and real-time inference demands. The proposed NES-RUP method, which selectively perturbs subsets of intermediate-layer feature channels using NES optimization and uniformly distributed noise, is explicitly designed to satisfy these operational constraints. NES operates in a black-box setting by estimating gradient directions using only forward-pass queries. At each iteration, perturbation samples are drawn, and variations in classification confidence are used to approximate the loss gradient. This eliminates the need for backpropagation or access to internal model parameters, making NES-RUP well-suited for real-world attack scenarios where gradient information is unavailable.

The computational cost of NES-RUP grows linearly with the number of sampled directions K and the number of selected feature channels N, both of which are tunable hyperparameters. The per-sample time complexity is $O(T \cdot K \cdot F_{\text{fwd}})$, where T is the maximum number of iterations, K is the number of NES samples per iteration, and F_{fwd} denotes the computational cost of a single forward pass through the downstream model partition. The additional arithmetic for masking and updating N channels adds $O(N \cdot H \cdot W)$ operations per iteration (with $H \times W$ denoting spatial size), which is negligible compared to F_{fwd}. Since intermediate feature maps are already computed as part of normal inference, the overhead of NES-RUP consists almost entirely of repeated forward passes, keeping the method lightweight. This design allows adversaries to balance attack strength and efficiency according to device capabilities or available time, while maintaining minimal added cost in practice. Furthermore, by perturbing only a small subset of feature channels, NES-RUP reduces communication overhead in horizontally partitioned models, a critical advantage in low-bandwidth or intermittently connected environments. The method requires no architectural modifications, auxiliary modules, or trainable parameters, enabling seamless integration into existing DDL pipelines and scalability to deeper architectures and larger deployment scenarios.

5. Experimental methodology

In this section, we describe the experimental setup, including datasets, models, and evaluation metrics, to rigorously assess the stealthiness and effectiveness of our proposed adversarial perturbation methods. We also use PseudoNet, a lightweight feature-map anomaly detector, conceptually inspired by [41], utilized to quantify the detectability of perturbations introduced at intermediate layers in horizontally distributed deep learning environments.

5.1. Detector implementation

We adopt PseudoNet, a lightweight anomaly detector motivated by the design in [41], to evaluate the stealthiness of adversarial perturbations objectively. This anomaly detector is selected for its computational efficiency, simplicity, and demonstrated ability to accurately approximate clean intermediate-layer activations. It requires no architectural modifications or significant computational overhead, making it ideally suited for AIoT and edge deployments where anomaly detection must operate in real time without disrupting the inference pipeline.

The anomaly detector is trained using intermediate feature maps collected from clean (unperturbed) samples during standard inference. Formally, given a clean feature map $F \in \mathbb{R}^{C\times H\times W}$, the proposed detector is optimized to produce an approximation $\hat{F}$ by minimizing the Mean Squared Error (MSE) loss:

$$\text{MSE}(F, \hat{F}) = \frac{1}{C \times H \times W} \sum_{c=1}^{C} \sum_{h=1}^{H} \sum_{w=1}^{W} (F_{c,h,w} - \hat{F}_{c,h,w})^2. \tag{6}$$

After training, an MSE threshold is empirically determined from the distribution of MSE values on a clean validation set. This threshold is defined statistically as:

$$\tau_{\text{MSE}} = \mu_{\text{MSE}} + 3 \times \sigma_{\text{MSE}}, \tag{7}$$

where μ_{MSE} and σ_{MSE} denote the mean and standard deviation of MSE values computed from clean validation samples. During evaluation, any feature map with an MSE above τ_{MSE} is flagged as adversarial, providing a quantitative measure of detectability.

The choice of PseudoNet is motivated by several key considerations. First, its simplicity and efficiency make it suitable for deployment in resource-constrained, distributed environments. Unlike complex detection frameworks, PseudoNet requires only lightweight offline training and does not impose architectural modifications on the underlying model. Second, it is model-agnostic and does not rely on prior knowledge of the adversarial strategy, making it broadly applicable across various attack methods. Lastly, the general concept of using lightweight surrogate networks for intermediate-layer anomaly detection has been explored in prior work, supporting its suitability as a reliable and objective baseline for evaluating the stealthiness of adversarial attacks. By adopting PseudoNet as the detection mechanism, we ensure a fair and reproducible evaluation of both random and NES-RUP optimized perturbation strategies in distributed deep learning settings.

5.2. Datasets and model architectures

To evaluate the proposed adversarial perturbation strategies, we utilize two widely adopted convolutional neural network architectures, VGG16 and ResNet50, selected for their architectural diversity and established use in adversarial robustness research. These models provide a representative spectrum of depth and structural complexity, enabling a robust assessment of attack efficacy under varying network

Table 1
Summary of key hyperparameters for NES-RUP and baseline methods.

Parameter	Value (VGG16/CIFAR-10)	Value (ResNet50/Mini-ImageNet)
Max iterations (T)	300	300
Step size (α)	0.4	0.4
Exploration scale (σ)	0.3	0.3
NES samples (K)	10	10
Perturbed channel ratio (N/C)	40%	40%
Stopping criterion	Misclassification or confidence $< P_{\text{th}}$	Same

configurations. In addition, to test generality beyond CNNs, we fine-tuned a DeiT-Tiny Vision Transformer on CIFAR-10 and partitioned it across four devices.

For the VGG16 experiments, we use the CIFAR-10 dataset, which contains 60,000 color images across 10 object categories. Owing to its moderate complexity, CIFAR-10 supports controlled and interpretable experimentation across different attack modalities. Standard preprocessing techniques, including normalization, random horizontal flipping, and random cropping, are applied to ensure training consistency and enhance generalization. ResNet50 experiments are conducted on the Mini-ImageNet dataset, which presents a more challenging classification task due to its higher intra-class variability and increased number of classes. Mini-ImageNet serves as a rigorous benchmark for evaluating adversarial performance under complex visual representations. Both models are deployed in a horizontally partitioned collaborative inference setting, where the network layers are distributed across multiple AIoT nodes. Specifically, VGG16 is partitioned across five nodes, while ResNet50 is divided among three. In accordance with the assumed threat model, the first and final nodes in each configuration are considered trusted, while adversarial perturbations are selectively introduced at intermediate nodes by manipulating transmitted feature maps between partitions. Both models are trained from scratch using standard optimization settings, including stochastic gradient descent with momentum, weight decay, and learning rate scheduling. After training convergence, the VGG16 model achieves a clean top-1 classification accuracy of 91.4% on the CIFAR-10 dataset, while the ResNet50 model attains 96.8% accuracy on the Mini-ImageNet dataset. These baseline results confirm that both models are well-trained and provide a reliable foundation for evaluating adversarial robustness in horizontally partitioned inference pipelines.

Perturbations are applied using the methodology described in Section 4. The perturbation magnitude (η) is empirically selected to balance attack strength and stealthiness. A classification confidence threshold (P_{th}) is used to define a successful misclassification, triggered when the model's confidence in the correct class drops below P_{th}. The number of perturbed channels (N) is tuned to minimize visibility in the feature space while ensuring reliable misclassification. For optimization-based attacks such as NES-RUP, hyperparameters including the number of samples (K), noise scale (σ), and learning rate (α) are selected through validation to optimize convergence speed, attack efficacy, and evasion of detection. The maximum number of iterations is set independently for NES-based and random perturbation methods to reflect their distinct convergence behaviors.

For all NES-based adversarial perturbation methods (NES-RUP and NES-random-Gaussian (NES-RG)), we fix the maximum number of iterations to 300, using a perturbation exploration scale $\sigma = 0.3$, a step size $\alpha = 0.4$, and $K = 10$ samples per gradient estimation. For random perturbation baselines (RUP and GNA), the attack proceeds by perturbing one randomly selected channel at a time until the success criterion is met. The same stopping condition is used for NES-based attacks: the attack terminates once the model misclassifies the sample or the confidence for the correct class falls below the threshold P_{th}. All attacks are evaluated under the hyperparameter settings summarized in Table 1. Moreover, it should be noted that in all methods, we adopt a baseline setting where 40% of the feature map channels are randomly selected at the start of the attack as the candidate set for perturbation. This choice provides a consistent benchmark across models and attack variants. However, this is not a strict assumption. In the results section, we further evaluate NES-RUP under smaller perturbation budgets (e.g., 10%–20% of channels), showing that the attack remains highly effective even when only a limited subset of channels can be manipulated.

5.3. Evaluation metrics and adversarial variants

To comprehensively evaluate the effectiveness and stealthiness of adversarial perturbations in distributed inference systems, we employ three core metrics: attack success rate, feature map distortion, and post-attack accuracy.

Attack success rate is defined as the percentage of test samples for which the adversarial perturbation leads to misclassification. A sample is considered successfully attacked if either (i) the model predicts an incorrect class, or (ii) the classification confidence for the correct class falls below a predefined threshold P_{th}. This dual criterion captures both direct prediction failure and confidence degradation, which are critical in AIoT applications where confidence thresholds may influence downstream decisions. *Feature map distortion* is computed as the ℓ_2-norm between clean and perturbed feature maps at the attacked partition. This metric quantifies the statistical deviation introduced at the internal representation level and serves as a proxy for detectability. In fact, larger distortions imply a higher likelihood of being flagged by anomaly-based defenses. *Post-attack accuracy* measures the overall degradation in model performance caused by adversarial perturbations. It is defined as the model's accuracy on perturbed inputs, reported alongside the baseline to show performance degradation. This metric reflects the end-to-end impact of adversarial manipulation on distributed inference pipelines.

Together, these metrics provide a multi-dimensional view of adversarial behavior, capturing both the effectiveness of attacks and their visibility to detection mechanisms. They enable fair and interpretable comparisons across perturbation strategies and noise configurations.

To evaluate these metrics, we consider four adversarial variants that differ in noise distribution and optimization strategy while maintaining consistent random channel selection. The first method, referred to as RUP, applies uniformly distributed noise to randomly selected intermediate-layer channels without any optimization. This serves as a baseline for unstructured, non-adaptive perturbations. The second variant, GNA, injects Gaussian-distributed noise into randomly selected channels. As a commonly used black-box baseline, GNA also lacks optimization and represents a standard non-adaptive threat model in feature space. The third method, NES-RUP, incorporates NES-based optimization while restricting perturbations to a randomly selected subset of channels. It uses uniform noise and uses gradient-free, query-efficient search to improve attack effectiveness without increasing the perturbation scope. The final variant, NES-RG, combines NES optimization with Gaussian-distributed perturbations over random channel subsets. This configuration enables a direct comparison between uniform and Gaussian noise distributions under the NES framework.

All NES-based variants use the two-sided gradient estimation defined in Eq. (4), where the direction vectors u_k are sampled from either a uniform or Gaussian distribution. Uniform noise introduces bounded, lower-variance perturbations that enhance stealth and reduce detectability, whereas Gaussian noise yields unbounded, higher-variance perturbations that may improve attack strength at the cost of increased visibility to anomaly detectors. These four variants allow a systematic analysis of how noise distribution (uniform vs. Gaussian) and optimization strategy (random vs. NES) influence attack efficacy, convergence behavior, and stealth. In all cases, the selection of channel subsets is performed randomly, without activation-based heuristics or importance sampling.

6. Results and analysis

In this section, we comprehensively evaluate the effectiveness and stealthiness of our proposed adversarial perturbation method, NES-RUP. Specifically, we demonstrate that selectively perturbing a subset of intermediate-layer channels using uniformly distributed noise, optimized via NES, significantly improves attack efficiency and reduces detectability compared to traditional random perturbation approaches. We analyze several key factors, including the impact of perturbation type (uniform versus Gaussian noise), the efficiency gains from targeting only a subset of feature channels, and the effectiveness of NES-based optimization in black-box adversarial settings. Additionally, we benchmark NES-RUP against GNA, a common baseline for black-box adversarial perturbation that injects indiscriminate Gaussian noise into feature maps, to highlight the advantages of structured and query-driven perturbations.

6.1. Effect of perturbation type and magnitude

This experiment evaluates the impact of two additive noise types, Gaussian and Uniform, on the robustness of horizontally partitioned deep neural networks under adversarial conditions. Perturbations are injected into intermediate feature maps at various magnitudes, with affected channels selected randomly. To ensure neutrality, the selection is non-prioritized and agnostic to activation scores or saliency metrics, allowing for an unbiased comparison of the intrinsic disruptive power of each noise distribution. This effectively corresponds to a comparative evaluation of the RUP and GNA methods introduced earlier, which differ only in their underlying noise distribution.

Fig. 3 presents results for the VGG16 model on CIFAR-10. Each row corresponds to an attack applied at a different device, and the plots report three key metrics: attack success rate, classification accuracy, and ℓ_2 distortion. Across all cases, Uniform noise consistently outperforms Gaussian noise, achieving higher attack success at lower perturbation magnitudes. For instance, Uniform noise surpasses 80% success at $\eta =$ 0.2–0.3, whereas Gaussian noise requires $\eta \geq 0.4$ to reach comparable performance. Notably, classification accuracy deteriorates more rapidly under Uniform noise, particularly at deeper layers (e.g., Device 4), where feature representations are more abstract and compact. However, this heightened impact is not due to greater distortion. As shown in the third column of Fig. 3, Gaussian noise induces significantly higher ℓ_2 deviations compared to Uniform noise. Similar trends are observed in ResNet50 experiments on Mini-ImageNet, where perturbations are introduced at Device 2. Fig. 4 shows that Uniform noise again achieves substantially higher success rates and sharper accuracy drops at lower magnitudes, despite inducing lower ℓ_2 distortion. At $\eta = 0.4$, Uniform noise exceeds 75% success while Gaussian noise remains under 50%.

In summary, uniformly distributed noise presents a more severe threat to horizontally partitioned CNN inference in the context of the two evaluated models. It achieves faster and more reliable confidence suppression while maintaining a lower distortion footprint, making it a strong candidate for stealthy adversarial strategies in distributed AIoT systems. A plausible explanation for this experiment lies in the statistical properties of the two noise types. Uniform perturbations are bounded within a fixed range, ensuring that altered feature values

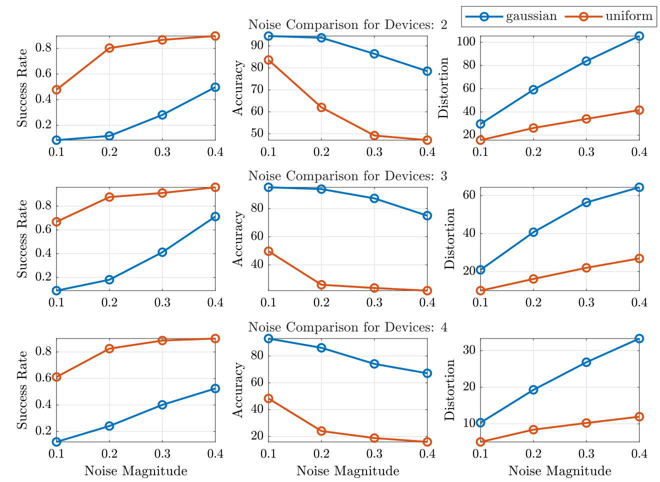

Fig. 3. Attack success rate, post-attack accuracy and average of feature map distortion on VGG16 (CIFAR-10) under Gaussian and uniform noise perturbations at varying magnitudes.

remain closer to their clean counterparts. This boundedness makes the resulting activations resemble the natural variability observed in intermediate feature maps, thereby maintaining a relatively low distortion footprint. In contrast, Gaussian perturbations are unbounded and may occasionally generate extreme outliers, which inflate ℓ_2 distortion and make the perturbations more conspicuous to detection mechanisms. As a result, RUP appears inherently more stealthy than GNA, even though its lack of optimization still limits its overall effectiveness compared to NES-RUP.

6.2. Analysis of NES-RUP method

This section presents a detailed evaluation of the NES-RUP method through two key analyses: first, by comparing its performance to non-optimized baselines and alternative perturbation distributions; and second, by examining its sensitivity to the number of perturbed channels in both VGG16 and ResNet50 models.

6.2.1. Comparison of optimized and non-optimized noise perturbations

This experiment evaluates the performance of both optimized (NES-RUP, NES-RG) and non-optimized (RUP, GNA) perturbation strategies on VGG16 and ResNet50 models, applied at intermediate devices in a horizontally partitioned inference setup. All methods use the same proportion of perturbed channels and share the stopping criteria defined in Section 5, ensuring a fair comparison across noise types and optimization strategies.

As shown in Fig. 5, NES-RUP achieves the highest attack success rate across all devices, exceeding 95% in most cases. RUP also performs strongly, achieving 80%–95% success without any optimization. In contrast, GNA and NES-RG yield noticeably lower success rates, particularly at shallower devices. For example, at Device 2, NES-RUP achieves 97% success, RUP achieves 85%, whereas GNA and NES-RG remain below 20%. Post-attack accuracy further confirms the superiority of NES-RUP. It reduces model accuracy to below 30% on all devices, outperforming even RUP, which leaves around 40%–60% accuracy. Both GNA and NES-RG result in minimal accuracy drop, reflecting their lower adversarial impact. In terms of stealth, NES-RUP and NES-RG maintain the lowest average distortion, with NES-RUP achieving the best overall stealth-to-success trade-off. Despite its high attack strength, NES-RUP produces less distortion than both GNA and RUP, indicating efficient perturbation targeting. NES-RG also maintains low distortion, but at the cost of significantly reduced attack effectiveness. This is likely

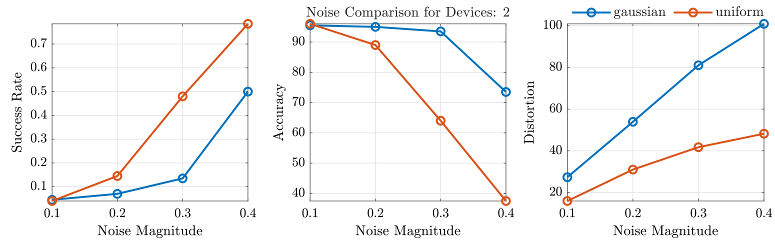

Fig. 4. Attack success rate, post-attack accuracy and average feature map distortion on ResNet50 (Mini-ImageNet) under Gaussian and uniform noise perturbations at varying magnitudes.

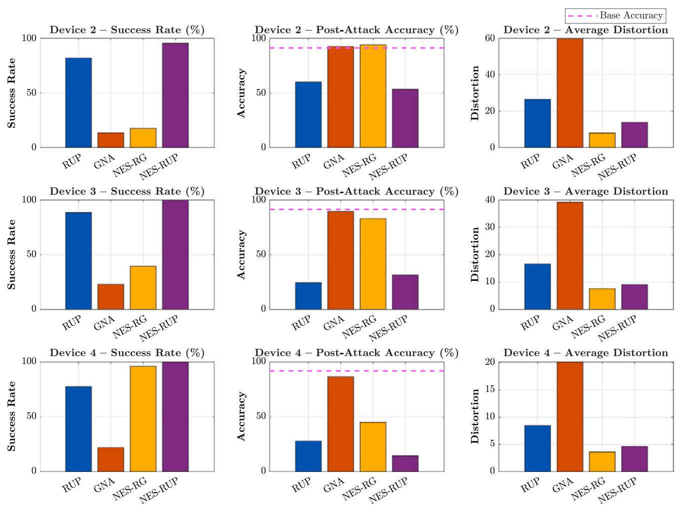

Fig. 5. Performance comparison of four adversarial perturbation methods (RUP, GNA, NES-RG, NES-RUP) on the VGG16 model across three partition points.

due to the high variance and unbounded nature of Gaussian perturbations, which can reduce the effectiveness of NES optimization by producing unstable gradient estimates. Fig. 6 presents the performance of RUP, GNA, NES-RG, and NES-RUP on Device 2 of the ResNet50 model. NES-RUP achieves the highest attack success rate and the largest drop in post-attack accuracy, while maintaining low average distortion. RUP follows closely with a strong success rate and a comparable post-attack accuracy, though it incurs higher distortion. In contrast, GNA and NES-RG exhibit lower success rates and leave significantly higher accuracy, with GNA producing the highest distortion overall. These results reaffirm the advantage of NES optimization and uniform noise, with NES-RUP achieving the best trade-off between effectiveness and stealth.

These results highlight that optimization with NES improves adversarial performance over random baselines. Moreover, uniform noise consistently outperforms Gaussian noise in both optimized and non-optimized settings, and finally NES-RUP delivers the best balance between adversarial perturbation strength and stealth, making it a strong adversarial candidate for real-world distributed inference attacks.

6.2.2. Impact of perturbed channel number on RUP and NES-RUP performance

To better understand the trade-off between attack strength and perturbation budget, we investigate how the number of perturbed feature map channels influences the effectiveness and stealthiness of RUP and NES-RUP. This analysis is conducted on both VGG16 and ResNet50 models under horizontally partitioned inference. By varying the number of perturbed channels across intermediate devices, we aim

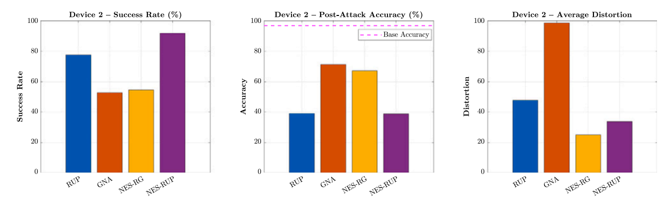

Fig. 6. Performance comparison of four adversarial perturbation methods (RUP, GNA, NES-RG, NES-RUP) on the Resnet50 model across three partition points.

to evaluate how each method scales in success rate, post-attack accuracy, and feature map distortion, and whether NES-based optimization remains effective under tighter perturbation budgets.

Fig. 7 analyzes how varying the number of perturbed channels affects the performance of RUP and NES-RUP across Devices 2, 3, and 4 of the VGG16 model. The results consistently show that NES-RUP achieves higher success rates and lower post-attack accuracy compared to RUP, even when perturbing fewer channels. This performance gap is most pronounced at lower perturbation levels, where NES-RUP maintains strong attack effectiveness while RUP struggles to converge. As the number of perturbed channels increases, RUP gradually approaches NES-RUP in success rate and impact, but this comes at the cost of significantly higher feature map distortion. In contrast, NES-RUP achieves high success rates with substantially lower distortion across all devices. For example, at Device 4, NES-RUP reaches near-maximal attack effectiveness while keeping distortion low, whereas RUP requires more than double the distortion to achieve comparable results.

Fig. 8 presents the effect of increasing the number of perturbed channels on the performance of RUP and NES-RUP at Device 2 in the ResNet50 model. NES-RUP consistently outperforms RUP across all metrics. Its success rate exceeds 90% even with 200 channels, while RUP requires nearly 500 channels to approach comparable effectiveness. This advantage is particularly evident at lower perturbation budgets, where NES-RUP maintains high attack success while inducing minimal distortion, preserving stealth under tighter channel budgets. Post-attack accuracy drops sharply for RUP as more channels are perturbed, but remains consistently lower for NES-RUP throughout. Distortion trends show that NES-RUP achieves strong adversarial impact with a much smaller distortion footprint, whereas RUP climbs steadily and exceeds 50 at maximum perturbation.

These findings confirm that NES-RUP achieves misclassification with fewer perturbed channels and lower distortion, making it both more efficient and more stealthy.

6.2.3. Dynamic split stress test

In practice, collaborative inference may reconfigure the partition point, referred to as the cut and defined as the boundary layer where the model is split across devices, at runtime due to load balancing, thermal limits, or link conditions [49]. To emulate this behavior, we perform a dynamic split stress test in which, for each input, the active cut is sampled at runtime from a small set of plausible split locations.

Table 2
Dynamic-split attack results on VGG16 (CIFAR-10) and ResNet50 (Mini-ImageNet).

Attack method	VGG16 (CIFAR-10)			ResNet50 (Mini-ImageNet)		
	Success (%)	Post-Acc (%)	Dist.	Success (%)	Post-Acc (%)	Dist.
RUP	79.0	28.0	13.06	49.0	62.0	68.48
GNA	22.0	88.0	29.48	29.0	91.0	129.68
NES-RUP	100.0	20.0	6.90	93.0	47.0	52.32

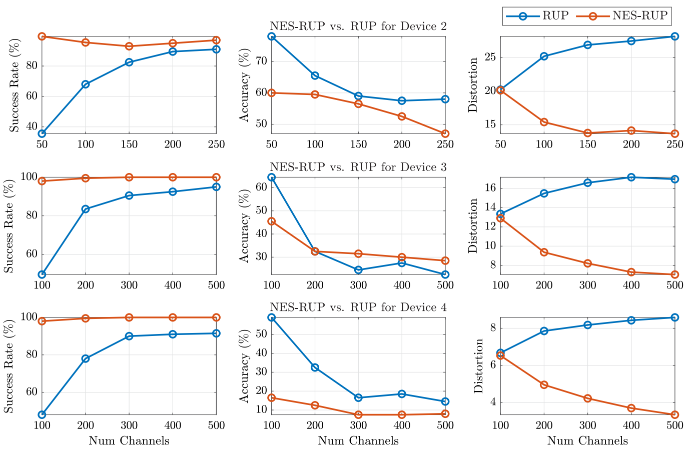

Fig. 7. Impact of varying the number of perturbed channels on RUP and NES-RUP performance across Devices 2–4 of VGG16.

Table 3
Adversarial perturbation performance on partitioned DeiT-Tiny (CIFAR-10). Metrics: success rate (%), distortion (ℓ_2), and post-attack accuracy (%).

Device	Method	Success (%)	Distortion	Post-Acc (%)
Device 2	RUP	72.0	405.8	84.0
	GNA	57.0	391.0	85.0
	NES-RUP	94.0	51.5	36.0
Device 3	RUP	60.0	376.3	86.0
	GNA	53.0	336.8	89.0
	NES-RUP	91.0	67.09	40.0

Concretely, for VGG16 on CIFAR-10 we draw the cut uniformly from Devices 3 and 4, and for ResNet-50 on Mini-ImageNet we sample between layer2 and layer3. For each input we compute the feature tensor at the sampled cut, inject the perturbation at the compromised device, and forward the result through the unchanged downstream model. We report success rate, mean steps to success, and mean ℓ_2 distortion measured on the attacked feature tensor, aggregated over the mixed cuts (here VGG16: n=500, ResNet-50: n=500), as summarized in Table 2.

6.2.4. Evaluation on Vision Transformers

Transformers have recently become central to computer vision, with Vision Transformers (ViT) and their data-efficient variants (DeiT) offering competitive accuracy compared to CNNs [50]. Unlike CNNs, ViTs operate on patch embeddings and rely on self-attention across tokens, yet in collaborative inference they are also split across devices at intermediate transformer blocks. This exposes their internal token embeddings to the same feature-space attack surface.

It should be noted that for CNNs, the perturbations are injected into intermediate feature maps ($C \times H \times W$) transmitted between devices. In contrast, Vision Transformers exchange sequences of token embeddings ($N \times D$), where N is the number of tokens and D is the embedding dimension, at the output of each transformer block. In the current experiments, adversarial perturbation is applied directly to these token embeddings at the partition boundary, which plays an equivalent role to feature maps in CNNs and thus constitutes the attack surface in transformer-based collaborative inference. To assess the generality of the proposed method, we fine-tuned a DeiT-Tiny (deit_tiny_patch16_224) model [50] on CIFAR-10 and partitioned it across four devices, achieving an accuracy of 95.2%, with each device executing a consecutive set of transformer blocks. We then injected perturbations at intermediate devices (Device 2 and Device 3) using RUP, GNA, and NES-RUP under the same black-box threat model. Table 3 reports the results, highlighting the comparison between Gaussian noise baselines and NES-RUP.

The results show that while RUP and GNA can occasionally reduce confidence or induce misclassification, they require much higher distortions and often leave most predictions intact. In contrast, NES-RUP consistently achieves very high attack success with substantially lower distortion and a marked drop in post-attack accuracy, demonstrating that even compact transformer architectures remain vulnerable when partitioned across devices. These findings confirm that NES-RUP is effective beyond CNNs, generalizing to transformer-based distributed inference.

6.3. Interpretability analysis

To complement the quantitative evaluation, we further examine how different perturbation strategies modify internal representations through qualitative feature-map and attention-based visualizations.

6.3.1. Qualitative visualization of feature-space perturbations

To qualitatively examine how different perturbation strategies alter internal representations in the VGG16 model, Figs. 9–11 visualize the clean feature map, the adversarially perturbed feature map, and the corresponding absolute difference $|\Delta|$ for the most-affected channel at the attacked device. The most-affected channel is defined as the channel exhibiting the largest ℓ_2 deviation between the clean and adversarial feature maps, providing an informative and representative view of the perturbation structure. These visualizations highlight the distinct characteristics of the RUP, GNA, and NES-RUP attacks and illustrate how NES-based optimization produces more targeted and lower-magnitude perturbations compared to non-optimized noise methods.

6.3.2. Attention-based interpretability analysis in transformer models

To better understand how different feature-space perturbation strategies affect the internal behavior of transformer models, we visualize in Fig. 12 the CLS-to-patch attention (where CLS denotes the classification token) of the last DeiT-Tiny block for a clean input image and for the same image after applying RUP, GNA, and NES-RUP attacks at Device 2. Each row shows the raw attention map produced under a given attack, together with an attention-difference heatmap that captures the signed change relative to the clean attention (red: increased attention, blue: decreased attention). The visualizations reveal a clear distinction between optimized and non-optimized perturbations. RUP and GNA lead to large, irregular, and high-magnitude deviations in attention, spreading across many spatial tokens and substantially altering the global attention distribution. In contrast, NES-RUP induces smaller, more spatially concentrated changes while largely preserving the overall attention structure of the clean model. This localized and low-magnitude behavior reflects the gradient-guided nature of NES optimization, which selectively perturbs the most influential channels rather than injecting broad random noise. These results provide an interpretable, model-level confirmation that NES-RUP produces more targeted and stealthy perturbations compared to classical randomized attacks.

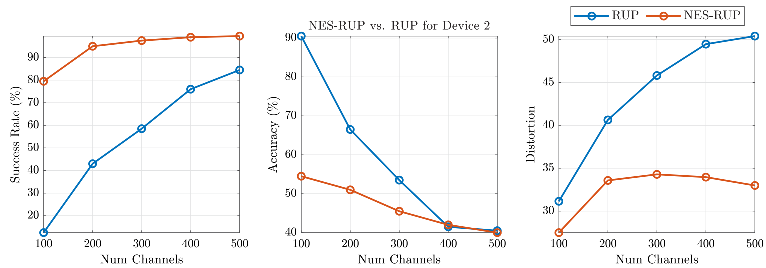

Fig. 8. Impact of varying the number of perturbed channels on RUP and NES-RUP performance at Device 2 of ResNet50.

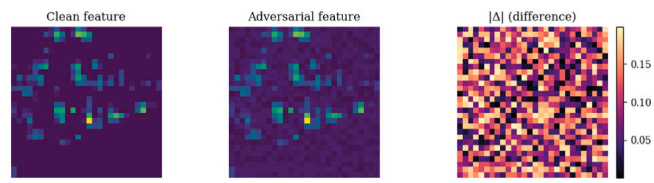

Fig. 9. Visualization of the clean feature map, adversarial feature map, and absolute difference $|\Delta|$ for the most-affected channel under the **RUP** attack.

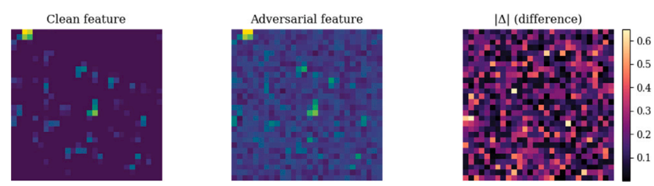

Fig. 10. Visualization of the clean feature map, adversarial feature map, and absolute difference $|\Delta|$ for the most-affected channel under the **GNA** attack.

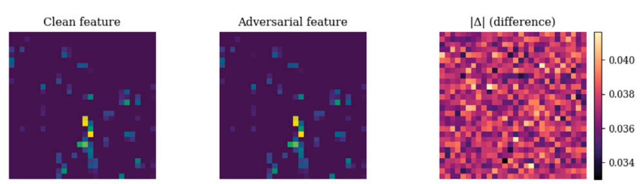

Fig. 11. Visualization of the clean feature map, adversarial feature map, and absolute difference $|\Delta|$ for the most-affected channel under the **NES-RUP** attack.

6.4. Evaluation of existing defense mechanisms

The resilience of distributed inference systems is next evaluated against two lightweight defenses proposed in prior work, namely the PseudoNet feature-map anomaly detector and low-pass filtering (LPF). PseudoNet approximates clean feature activations and flags anomalous deviations using a reconstruction-error threshold, while LPF attempts to attenuate injected noise by smoothing intermediate representations.

6.4.1. PseudoNet-based detection of feature-space attacks

To assess the detectability of adversarial perturbations introduced in the intermediate feature space, we adapt a lightweight anomaly detector, referred to as PseudoNet, based on the approach in [41]. This module is a compact convolutional neural network trained to approximate the output of an untrusted node using only the features from a preceding trusted device. The intuition is that, under clean conditions, PseudoNet produces accurate reconstructions of unperturbed feature maps. However, when adversarial perturbations are injected, the reconstruction error, measured via MSE, is expected to exceed a threshold, triggering anomaly detection. Figs. 13 and 14 show the MSE distributions computed on clean inputs for VGG16 and ResNet50, respectively. Thresholds are computed as Eq. (7), defined as $\tau_{\mathrm{MSE}} = \mu_{\mathrm{MSE}} + 3\sigma_{\mathrm{MSE}}$, yielding 0.0181 for VGG16 and 0.0114 for ResNet50. The tighter distribution in ResNet50 indicates more stable intermediate representations compared to VGG16. At inference time, the system computes the MSE between the actual feature map from the untrusted node and the output generated by PseudoNet. If the MSE exceeds the precomputed threshold, an attack is flagged. Detection performance is evaluated for three perturbation methods of RUP, GNA, and the proposed NES-RUP.

Table 4 summarizes the detection rate, attack success rate, post-attack accuracy, and average distortion across both models. On ResNet50, PseudoNet detects GNA with high reliability (96.4%), while detection performance for RUP is moderate at best (54.4%), reflecting limited sensitivity to unoptimized noise. Moreover, it struggles significantly with NES-RUP, detecting only 29.4% of successful attacks despite their structured nature and low distortion. On VGG16, detection rates are generally lower, with only 10.0% of RUP and 7.5% of NES-RUP cases flagged. Despite a 86.0% attack success rate for NES-RUP on VGG16, PseudoNet detects merely 7.5% of those cases, illustrating a significant blind spot. The low distortion of NES-RUP across both models further contributes to its ability to evade detection by PseudoNet.

These results demonstrate that while PseudoNet performs well for high-distortion or stochastic attacks like GNA, it is insufficient against low-distortion, optimized attacks. Particularly in VGG16, which has less feature stability than ResNet50, the detector is more easily fooled by structured perturbations. This highlights the challenge of designing lightweight detection mechanisms that can generalize across architectures and detect adversarial behaviors that closely mimic natural feature activations.

6.4.2. Resilience against low-pass filtering mitigation

To evaluate the efficacy of LPF presented in [51] as a lightweight mitigation strategy against adversarial perturbations, we conduct experiments under both clean and adversarial conditions using horizontally partitioned VGG16 and ResNet50 models. In each configuration, LPF is applied immediately after the intermediate device that is potentially untrusted or subject to adversarial perturbation (e.g., Device 2 or Device 3). This placement reflects a realistic defense scenario, where the LPF operates downstream of a compromised node in an attempt to suppress adversarial noise before it propagates through the remainder of the inference pipeline. The objective is to assess whether such post-perturbation filtering can attenuate adversarial effects while maintaining acceptable classification performance.

The LPF considered in this study is implemented as a blended average pooling operation, designed to smooth the internal feature representations by combining each feature map with its local average. Formally, the filtered feature map F_{LPF} is computed as:

$$F_{\mathrm{LPF}} = \beta \cdot \mathrm{AvgPool}(F) + (1 - \beta) \cdot F, \tag{8}$$

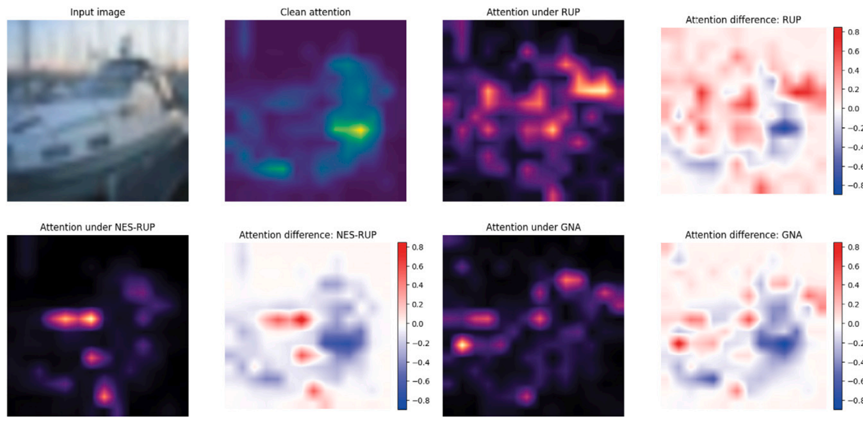

Fig. 12. Attention-based visualization under RUP, GNA, and NES-RUP attacks.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Detection performance of PseudoNet across attack types on VGG16 (CIFAR-10) and ResNet50 (Mini-ImageNet). Lower distortion correlates with reduced detectability.

Attack method	VGG16 (CIFAR-10)				ResNet50 (Mini-ImageNet)			
	Det. rate (%)	Success (%)	Post-Acc (%)	Dist.	Det. rate (%)	Success (%)	Post-Acc (%)	Dist.
RUP	10.0	66.0	62.5	25.27	54.4	16.5	85.9	30.75
GNA	79.5	16.0	82.0	51.21	96.4	5.5	91.9	54.71
NES-RUP	7.5	86.0	59.0	13.68	29.4	45.6	75.3	13.76

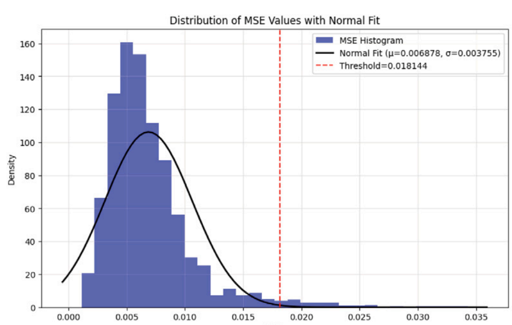

Fig. 13. MSE distribution on clean CIFAR-10 inputs (VGG16).

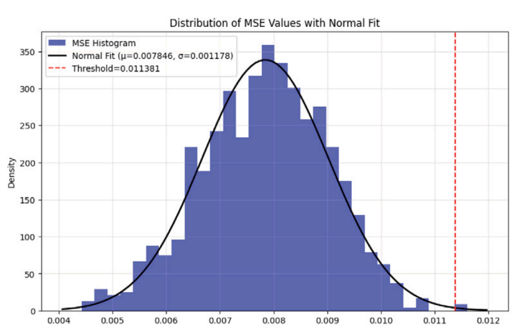

Fig. 14. MSE distribution on clean Mini-ImageNet inputs (ResNet50).

where β denotes the blending factor and $\mathtt{AvgPool}$ represents a kernel-based average pooling operation. This filtering is applied to the attacked device after perturbation injection and serves to attenuate high-frequency components that are often exploited in noise-based adversarial attacks. Table 5 reports the clean classification accuracy of the

Table 5
Baseline accuracy of VGG16 and ResNet50 under LPF and no-LPF conditions.

Model	No LPF	With LPF (Device 3)
VGG16 (CIFAR-10)	91.4%	89.6%
ResNet50 (Mini-ImageNet)	96.8%	95.2%

Table 6
Post-attack accuracy (with LPF applied at attacked device) for different adversarial methods.

Attack method	VGG16 (CIFAR-10)		ResNet50 (Mini-ImageNet)	
	Post-Acc (%)	Distortion	Post-Acc (%)	Distortion
RUP	58.20	25.67	40.00	47.79
GNA	86.40	55.63	93.00	107.89
NES-RUP	51.20	13.07	46.50	20.97

models with and without LPF applied. As expected, the smoothing effect of LPF introduces a slight degradation in baseline performance.

Table 6 presents the post-attack classification accuracy and corresponding distortion values when LPF is applied after the perturbed device. Across both architectures, the results show that LPF offers limited robustness against unstructured perturbations such as GNA and RUP. While there is some improvement in classification accuracy compared to their expected no-defense performance, substantial performance degradation remains, particularly for RUP, where accuracy drops to 58.2% on VGG16 and 40.0% on ResNet50. This indicates that LPF is only partially effective in mitigating high-magnitude, non-optimized perturbations and cannot reliably preserve model predictions under such attacks.

Moreover, the NES-RUP method demonstrates strong resilience to LPF, maintaining the lowest post-attack accuracy across both models despite introducing significantly lower distortion. This suggests that NES-RUP is more effective at evading LPF-based suppression due to its structure-aware optimization, which produces perturbations that closely mimic natural feature statistics.

These findings illustrate the limitations of LPF as a defense mechanism in distributed inference. While LPF can attenuate the impact of unstructured attacks, it remains largely ineffective against adaptive, structure-aware perturbations such as those generated by NES-RUP. Furthermore, the marginal reduction in clean accuracy introduced by LPF highlights the trade-off between robustness and fidelity in feature-space smoothing. Overall, the results emphasize the necessity of developing more advanced defense strategies that can reliably detect and neutralize low-distortion perturbations in collaborative inference systems.

6.4.3. Analysis of defense limitations and future directions

The results from the previous section show that current lightweight defenses are insufficient against NES-RUP. Although both LPF and PseudoNet anomaly detection provide partial protection against unstructured or high-distortion attacks, they struggle against low-distortion, structure-aware perturbations. Table 4 shows that PseudoNet detects stochastic Gaussian perturbations reliably but struggles to flag low-magnitude, uniformly distributed perturbations crafted by NES-RUP. Because these perturbations remain statistically close to natural activations, the reconstruction error often stays below the detection threshold, allowing the attack to evade detection. As shown in Tables 5 and 6, LPF not only fails to suppress NES-RUP but also reduces clean accuracy by nearly 2% on VGG16 and 1.6% on ResNet50. This illustrates an inherent trade-off between robustness and fidelity: while LPF can attenuate high-frequency Gaussian noise, it cannot counter perturbations that closely mimic natural feature statistics, and it simultaneously degrades baseline performance.

While the design of robust countermeasures is beyond the scope of this work, our findings indicate that effective defenses may require more expressive statistical modeling of feature-map distributions beyond simple MSE, stronger semantic consistency checks to enforce agreement between intermediate activations and outputs, and ensemble-style detection across multiple nodes to reduce single-point failures. Another promising direction is adversarial training directly in the feature space using query-efficient perturbations similar to NES-RUP. Together, these directions highlight opportunities for future research aimed at mitigating covert feature-space adversarial attacks in distributed inference.

7. Conclusion

This paper presents a query-efficient and stealthy adversarial perturbation method, termed NES-RUP, targeting intermediate feature maps in horizontally partitioned convolutional neural networks. Using Natural Evolution Strategies (NES), the attack selectively perturbs a small subset of feature channels using uniform noise, optimized to induce misclassification while maintaining low distortion. Experimental results on VGG16 and ResNet50 demonstrate that NES-RUP achieves high attack success rates with minimal perturbation, outperforming unoptimized baselines such as Gaussian Noise Attack (GNA) and Random Uniform Perturbation (RUP). Moreover, we confirmed its effectiveness on transformer-based distributed inference, demonstrating generality across architectures. Qualitative visualizations and attention-based interpretability analysis further confirm that NES-RUP alters internal feature representations in a structured yet low-distortion manner. In addition to its effectiveness, NES-RUP is shown to be resilient against lightweight defenses commonly applied in distributed inference systems. Specifically, the method evades low-pass filtering (LPF) and a reconstruction-based anomaly detector (PseudoNet), both of which exhibit limited ability to suppress or detect the optimized, structure-aware perturbations generated by NES-RUP. These findings highlight the emerging threat of feature-space adversarial attacks in collaborative deep learning pipelines and underscore the limitations of existing lightweight defenses. Future work will explore adaptive defense mechanisms that integrate statistical feature modeling and semantic consistency checks to better detect and mitigate low-distortion adversarial behavior in partitioned inference settings.

CRediT authorship contribution statement

Arash Golabi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Abdelkarim Erradi:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Funding acquisition. **Ahmed Bensaid:** Visualization, Validation, Supervision, Project administration. **Abdulla Al-Ali:** Visualization, Supervision, Project administration, Funding acquisition. **Uvais Qidwai:** Visualization, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] R. Zhang, H. Jiang, J. Geng, F. Tian, Y. Ma, H. Wang, A high-performance dataflow-centric optimization framework for deep learning inference on the edge, J. Syst. Archit. 152 (2024) 103180.
[2] Y. Sun, H. Ochiai, H. Esaki, Decentralized deep learning for multi-access edge computing: A survey on communication efficiency and trustworthiness, IEEE Trans. Artif. Intell. 3 (6) (2022) 963–972.
[3] Z. Wang, M. Wen, Y. Xu, Y. Zhou, J.H. Wang, L. Zhang, Communication compression techniques in distributed deep learning: A survey, J. Syst. Archit. 142 (2023) 102927.
[4] X. Wang, Y. Han, V.C.M. Leung, D. Niyato, X. Yan, X. Chen, Convergence of edge computing and deep learning: A comprehensive survey, IEEE Commun. Surv. Tutor. 22 (2) (2020) 869–904.
[5] X. Shen, X. Luo, F. yuan, B. Wang, Y. Chen, D. Tang, L. Gao, Privacy-preserving multi-party deep learning based on homomorphic proxy re-encryption, J. Syst. Archit. 144 (2023) 102983.
[6] E. Baccour, N. Mhaisen, A.A. Abdellatif, A. Erbad, A. Mohamed, M. Hamdi, M. Guizani, Pervasive AI for IoT Applications: A Survey on Resource-Efficient Distributed Artificial Intelligence, vol. 24, IEEE, 2022, pp. 2366–2418, no. 4. arXiv:2105.01798.
[7] F. Dong, X. Ge, Q. Li, J. Zhang, D. Shen, S. Liu, X. Liu, G. Li, F. Wu, J. Luo, PADP-FedMeta: A personalized and adaptive differentially private federated meta learning mechanism for AIoT, J. Syst. Archit. 134 (2023) 102754.
[8] I.J. Goodfellow, J. Shlens, C. Szegedy, Explaining and harnessing adversarial examples, 2014, arXiv preprint arXiv:1412.6572.
[9] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, A. Vladu, Towards deep learning models resistant to adversarial attacks, 2017, arXiv preprint arXiv:1706.06083.
[10] N. Carlini, D. Wagner, Towards evaluating the robustness of neural networks, in: 2017 IEEE Symposium on Security and Privacy, SP, 2017, pp. 39–57.
[11] S.-M. MoosaviDezfooli, A. Fawzi, P. Frossard, DeepFool: A simple and accurate method to fool deep neural networks, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2016, pp. 2574–2582.
[12] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z.B. Celik, A. Swami, The limitations of deep learning in adversarial settings, in: 2016 IEEE European Symposium on Security and Privacy, EuroS&P, 2016, pp. 372–387.
[13] A. Modas, S.-M. Moosavi-Dezfooli, P. Frossard, SparseFool: A few pixels make a big difference, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2019, pp. 9079–9088.
[14] S. Bhambri, S. Muku, A. Tulasi, A.B. Buduru, A survey of black-box adversarial attacks on computer vision models, 2019, arXiv preprint arXiv:1912.01667.
[15] T. Salimans, J. Ho, X. Chen, S. Sidor, I. Sutskever, Evolution strategies as a scalable alternative to reinforcement learning, 2017, arXiv preprint arXiv:1703.03864.
[16] T. Ben-Nun, T. Hoefler, Demystifying parallel and distributed deep learning, ACM Comput. Surv. 52 (2018) 1–43.
[17] M. Langer, Z. He, W. Rahayu, Y. Xue, Distributed training of deep learning models: A taxonomic perspective, IEEE Trans. Parallel Distrib. Syst. 31 (2020) 2802–2818.

[18] A. Castelló, M.F. Dolz, E.S. Quintana-Ortí, J. Duato, Analysis of model parallelism for distributed neural networks, in: Proceedings of the 26th European MPI Users' Group Meeting, ACM, New York, NY, USA, 2019, pp. 1–10.

[19] J. Chen, X. Ran, Deep learning with edge computing: A review, Proc. IEEE 107 (8) (2019) 1655–1674.

[20] C. Hu, B. Li, Distributed inference with deep learning models across heterogeneous edge devices, in: IEEE INFOCOM 2022 - IEEE Conference on Computer Communications, 2022, pp. 330–339.

[21] R. Stahl, Z. Zhao, D. Mueller-Gritschneder, A. Gerstlauer, U. Schlichtmann, Fully distributed deep learning inference on resource-constrained edge devices, in: D.N. Pnevmatikatos, M. Pelcat, M. Jung (Eds.), Embedded Computer Systems: Architectures, Modeling, and Simulation, Springer International Publishing, Cham, 2019, pp. 77–90.

[22] Y. Chen, T. Luo, W. Fang, N.N. Xiong, EdgeCI: Distributed workload assignment and model partitioning for cnn inference on edge clusters, ACM Trans. Internet Technol. 24 (2) (2024) 1–24.

[23] R. Stahl, A. Hoffman, D. Mueller-Gritschneder, A. Gerstlauer, U. Schlichtmann, DeeperThings: Fully distributed CNN inference on resource-constrained edge devices, Int. J. Parallel Program. 49 (2021) 600–624.

[24] O. Kayode, A. Tosun, Deep Q-network for enhanced data privacy and security of IoT traffic, in: 2020 IEEE 6th World Forum on Internet of Things, WF-IoT, 2020, pp. 1–6.

[25] Y. Chen, S. Kar, J.M. Moura, The internet of things: Secure distributed inference, IEEE Signal Process. Mag. 35 (5) (2018) 64–75, arXiv:1806.09199.

[26] A. Ilyas, L. Engstrom, A. Athalye, J. Lin, Black-box adversarial attacks with limited queries and information, in: International Conference on Machine Learning, PMLR, 2018, pp. 2137–2146.

[27] A.A. Adeyemo, J.J. Sanderson, T.A. Odetola, F. Khalid, S.R. Hasan, StAIn: Stealthy avenues of attacks on horizontally collaborated convolutional neural network inference and their mitigation, IEEE Access 11 (January) (2023) 10520–10534.

[28] S. Pal, S. Rahman, M. Beheshti, A. Habib, Z. Jadidi, C.K. Karmakar, The impact of simultaneous adversarial attacks on robustness of medical image analysis, IEEE Access 12 (2024) 66478–66494.

[29] M. Fan, C. Chen, C. Wang, W. Zhou, J. Huang, On the robustness of split learning against adversarial attacks, Frontiers Artificial Intelligence Appl. 372 (2023) 668–675, arXiv:2307.07916.

[30] G. De La Torre Parra, P. Rad, K.K.R. Choo, N. Beebe, Detecting internet of things attacks using distributed deep learning, J. Netw. Comput. Appl. 163 (February) (2020).

[31] A. Adeyemo, P. Patel, S.R. Hasan, S. Member, Securing pseudo-model parallelism-based collaborative DNN inference for edge devices, IEEE Access PP (2024) 1.

[32] G. Rossolini, T. Baldi, A. Biondi, G. Buttazzo, Edge-only universal adversarial attacks in distributed learning, 2024, arXiv preprint arXiv:2411.10500.

[33] A. Uprety, D.B. Rawat, Mitigating poisoning attack in federated learning, in: 2021 IEEE Symposium Series on Computational Intelligence, SSCI, IEEE, 2021, pp. 01–07.

[34] T. Muhr, W. Zhang, Privacy-preserving detection of poisoning attacks in federated learning, in: 2022 19th Annual International Conference on Privacy, Security and Trust, PST 2022, IEEE, 2022, pp. 1–10, no. 1.

[35] H. Zhang, J. Sun, K.C. Tan, Z. Xu, Learning adaptive differential evolution by natural evolution strategies, IEEE Trans. Emerg. Top. Comput. Intell. 7 (3) (2023) 872–886.

[36] Y. Li, L. Li, L. Wang, T. Zhang, B. Gong, Nattack: Learning the distributions of adversarial examples for an improved black-box attack on deep neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 3866–3876.

[37] H. Qiu, L.L. Custode, G. Iacca, Black-box adversarial attacks using evolution strategies, in: Proceedings of the Genetic and Evolutionary Computation Conference Companion, ACM, New York, NY, USA, 2021, pp. 1827–1833.

[38] E.M. Campos, A. Gonzalez-Vidal, J.L. Hernández-Ramos, A. Skarmeta, FedRDF: A robust and dynamic aggregation function against poisoning attacks in federated learning, IEEE Trans. Emerg. Top. Comput. 13 (1) (2025) 48–67.

[39] T. Wang, Z. Zheng, F. Lin, Federated learning framew ork based on trimmed mean aggregation rules, SSRN Electron. J. (2022).

[40] L. Yao, S. Shao, S. Hariri, Resilient machine learning (rML) against adversarial attacks on industrial control systems, in: Proceedings of IEEE/ACS International Conference on Computer Systems and Applications, AICCSA, IEEE, 2023, pp. 1–8.

[41] M. Asif, M.K. Kazmi, M.A. Rahman, S.R. Hasan, S. Homsi, SHEATH: Defending Horizontal Collaboration for Distributed CNNs against Adversarial Noise, vol. 0109, 2024, arXiv:2409.17279.

[42] K. Ren, T. Zheng, Z. Qin, X. Liu, Adversarial attacks and defenses in deep learning, Engineering 6 (3) (2020) 346–360.

[43] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.

[44] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.

[45] A. Golabi, A. Erradi, A. Bensaid, A. Al-Ali, U. Qidwai, A dual-stage deep learning approach for robust detection and identification of hardware trojans using monte-carlo dropout, Int. J. Inf. Secur. 24 (3) (2025) 142.

[46] T.A. Odetola, S.R. Hasan, SoWaF: Shuffling of weights and feature maps: A novel hardware intrinsic attack (HIA) on convolutional neural network (CNN), in: Proceedings - IEEE International Symposium on Circuits and Systems, vol. 2021-May, 2021.

[47] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, R. Fergus, Intriguing properties of neural networks, 2013, arXiv preprint arXiv:1312.6199.

[48] A. Fawzi, O. Fawzi, P. Frossard, Analysis of classifiers' robustness to adversarial perturbations, Mach. Learn. 107 (3) (2018) 481–508.

[49] L. Yang, X. Shen, C. Zhong, Y. Liao, On-demand inference acceleration for directed acyclic graph neural networks over edge-cloud collaboration, J. Parallel Distrib. Comput. 171 (2023) 79–87.

[50] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jégou, Training data-efficient image transformers & distillation through attention, in: International Conference on Machine Learning, PMLR, 2021, pp. 10347–10357.

[51] A.A. Adeyemo, S.R. Hasan, Enhancing the security of collaborative deep neural networks: An examination of the effect of low pass filters, in: Proceedings of the ACM Great Lakes Symposium on VLSI, GLSVLSI, 2023, pp. 461–465.